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The equation $x^6 + xy + 2y^3 + 1 = 0$ has no integer solutions

Size $h = 85$, length $l = 12$

Theorem 1. *The equation*

$$x^6 + xy + 2y^3 + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + V_k^2$ on the curve (1), and the fact that an odd prime p with $(-1/p) = -1$ divides $U_k^2 + V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm form $a^2 + b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-1/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -(bc)^2 \equiv -1 \pmod{p}$, so -1 is a square modulo p . As $(-1/p) \neq 0$, it is a non-zero square, so $(-1/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + b^2 \neq 0$, and $v_p(a^2 + b^2)$ is even for every odd prime p with $(-1/p) = -1$.*

Proof. Clearly $a^2 + b^2 > 0$. Let p be an odd prime with $(-1/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + b'^2$ by Lemma 2. Since $a^2 + b^2 = p^{2k}(a'^2 + b'^2)$, we get $v_p(a^2 + b^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4}$. Then $(-1/n') = (-1/n)$.*

Proof. Write $-1 = \varepsilon 2^e q$ with $\varepsilon = -1$, $e = 0$ and $q = 1$. By multiplicativity and the reciprocity laws,

$$\left(\frac{-1}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since 4 is divisible by $4q$, the claim follows. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides 2, both $2s$ and $4s$ divide m , u is odd, $(-1/u) = -1$, and $n \equiv su \pmod{m}$. Suppose also that $n > 0$. Then there is an odd prime p with $(-1/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by 4, n' is odd and $n' \equiv u \pmod{4}$. Since $n > 0$, Lemma 4 gives $(-1/n') = (-1/u) = -1$. By multiplicativity, the product of $(-1/p)^{v_p(n')}$ over the prime factors p of n' is -1 , so some prime factor p of n' satisfies $(-1/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^6 + xy + 2y^3 + 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= 2x^2 - 7x - y + 6, \\ T_1 &= 136x^4 - 952x^3 + 68x^2y + 2482x^2 - 238xy - 2839x + 34y^2 + 204y + 1224, \\ U_1 &= 17x^3 - 84x^2 + 116x - 56, \\ V_1 &= -38x^2 + 97x - 65, \\ G_1 &= -17, \\ A_1 &= 53998x - 94403, \\ B_1 &= 24157x^2 - 99933x + 77103, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= 2x^2 - y, \\ T_2 &= 136x^4 + 68x^2y + 17x + 34y^2, \\ U_2 &= 17x^3 + 1, \\ V_2 &= -4, \\ G_2 &= -17, \\ A_2 &= 0, \\ B_2 &= -1, \end{aligned}$$

and, for the signs, we also use

$$\begin{aligned} Q_1 &= 8x^4 - 56x^3 + 4x^2y + 146x^2 - 14xy - 167x + 2y^2 + 12y + 72, \\ M_1 &= 17x^6 - 168x^5 + 732x^4 - 1692x^3 + 2189x^2 - 1506x + 433, \\ E_1 &= 48x^4 - 336x^3 + 876x^2 - 1000x + 432, \\ Q_2 &= 8x^4 + 4x^2y + x + 2y^2, \\ M_2 &= 17x^6 + 2x^3 + 1, \end{aligned}$$

$$E_2 = 48x^4 + 8x.$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 17 \cdot 19 \cdot 23 \cdot 37, \quad (4)$$

$$A_2U_2 + B_2V_2 = 2^2, \quad (5)$$

$$-F = Q_1H_1 - M_1, \quad (6)$$

$$8Q_1 = (4x^2 - 14x + 4y + 12)^2 + E_1. \quad (7)$$

$$-F = Q_2H_2 - M_2, \quad (8)$$

$$8Q_2 = (4x^2 + 4y)^2 + E_2. \quad (9)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + V_k^2 \quad (k = 1, 2), \quad (10)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 128^2 residue pairs modulo 128, we find that (1) has exactly 64 solutions modulo 128, and that at each of them $H_1 \bmod 128$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 128$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-1/u) = -1$; here for $s = 1, 2s = 2$ and $4s = 4$ divide 128. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-1/u) = -1$; here for $s = 16$, every prime factor of s divides 2, and $2s = 32$ and $4s = 64$ divide 128, and for $s = 32$, every prime factor of s divides 2, and $2s = 64$ and $4s = 128$ divide 128.

Step 2: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (6) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $0 \leq x \leq 1$. For $x = 0$ and $x = 1$ we have $M_1 > 0$ and $E_1 > 0$. Hence $M_1 > 0$ and $E_1 > 0$, so (7) gives $8Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (8) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $x = 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (9) gives $8Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

Step 3: primes p with $(-1/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-1/p) = -1$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (10) and Lemma 3 show that $v_p(H_kT_k) = v_p(U_k^2 + V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + V_k^2$ by (10), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_kU_k + B_kV_k$. For $k = 1$, we have $R_1 = 17 \cdot 19 \cdot 23 \cdot 37$. As p is odd and $p \neq 17$ and $p \neq 37$ because $(-1/17) = 1$ and $(-1/37) = 1$, we get $p \in \{19, 23\}$.

Checking the 19^2 residue pairs modulo 19 and the 23^2 residue pairs modulo 23, we find that F , H_1 and T_1 have no common zero modulo 19 and 23, a contradiction. For $k = 2$, we have $R_2 = 2^2$. But p is odd, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{128}$ with s and u as in Lemma 5, and $H_k > 0$ by Step 2. Hence Lemma 5 applies to $n = H_k$, so there is an odd prime p with $(-1/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 128 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
3	27	51	75	99	123
7	31	55	79	103	127
11	35	59	83	107	
15	39	63	87	111	
19	43	67	91	115	
23	47	71	95	119	
$\mathcal{R}_2$, the residues of H_2					
$48 = 16 \cdot 3$	$96 = 32 \cdot 3$	$112 = 16 \cdot 7$			